

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 2

- (1) Can you use the argument from problem 2 on Homework 1 to show that there are infinitely many primes that are congruent to 1 mod 4? What goes wrong?

Solution. The proof method doesn't work: Writing down a number that is congruent to 1 mod 4 does not ensure that one of its prime factors is congruent to 1 mod 4, since it could be a product of an *even* number of primes that are congruent to 3 mod 4.

Nevertheless, the conclusion is still true: it requires a somewhat different argument that needs a bit more elementary number theory.

- (2) Use strong induction to prove the following fact: Every integer $n \geq 12$ can be written in the form $n = 4a + 5b$ for $a, b \in \mathbb{Z}_{\geq 0}$, the set of non-negative integers.

Hint: You might need multiple base cases.

Solution. Let us consider the base cases $n = 12, 13, 14, 15$: For each of these we can find a, b explicitly:

$$12 = 4 \cdot 3 + 5 \cdot 0; \quad 13 = 4 \cdot 2 + 5 \cdot 1; \quad 14 = 4 \cdot 1 + 5 \cdot 2; \quad 15 = 4 \cdot 0 + 5 \cdot 3.$$

Now, for $n \geq 16$, we make the following inductive hypothesis: Every integer $12 \leq k < n$ can be written in the form $k = 4a + 5b$ with a, b non-negative integers. Moreover, we can assume that $n \geq 16$, so that $n - 4 \geq 12$.

We apply the hypothesis to $k = n - 4$: If $n - 4 = 4a + 5b$, then we see that $n = 4(a + 1) + 5b$. This shows that the statement is also true for n .

- (3) Suppose that F is a field. Verify the following facts using only the defining conditions for a field given in Lecture 4:

- (a) For any $a \in F$, $0 \cdot a = 0$.

Solution. We see via the distributive property that

$$0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a.$$

Adding $-0 \cdot a$ to both sides (inverse for addition) and using associativity for addition now gives us

$$0 = 0 \cdot a + (-0 \cdot a) = (0 \cdot a + 0 \cdot a) + (-0 \cdot a) = 0 \cdot a + (0 \cdot a + (-0 \cdot a)) = 0 \cdot a + 0 = 0 \cdot a.$$

- (b) For any $a \in F$, $(-1) \cdot a = -a$.

Solution.

$$0 = 0 \cdot a = (1 + (-1)) \cdot a = 1 \cdot a + (-1) \cdot a = a + (-1) \cdot a.$$

Here, in the first equality we have used what we showed in (a). This shows that $(-1) \cdot a = -a$ is the inverse of a for addition.

- (c) If $a, b \in F$ are such that $a \cdot b = 0$, then either $a = 0$ or $b = 0$.

Solution. This will use the existence of inverses for multiplication. If $b \neq 0$, then we can multiply the equality $0 = a \cdot b$ by b^{-1} and use associativity and property (a):

$$0 = (a \cdot b) \cdot b^{-1} = a \cdot (b \cdot b^{-1}) = a \cdot 1 = a.$$

- (4) Solve the following linear system of equations in the field $\mathbb{Z}/5\mathbb{Z}$:

$$\begin{aligned} 2x + 3y + z &= 1 \\ 4x + 2y + 3z &= 2 \end{aligned}$$

Solution. Extracting coefficients gives us the matrix representation

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 4 & 2 & 3 & 2 \end{array} \right]$$

First, we can make the top left entry 1 by multiplying the first row by $2^{-1} = 3 \in \mathbb{Z}/5\mathbb{Z}$:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 3 \\ 4 & 2 & 3 & 2 \end{array} \right]$$

Next we can add the first row to the second:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 3 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

Finally we can add the new second row to the first:

$$\left[\begin{array}{ccc|c} 1 & 0 & 4 & 3 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

This corresponds to the system of equations

$$\begin{aligned} x + 4z &= 3 \\ y + z &= 0 \end{aligned}$$

Therefore, we see that z can be given any arbitrary value in $\mathbb{Z}/5\mathbb{Z}$, and this determines the values of $x = 3 - 4z$ and $y = -z$. Since there are 5 possibilities for z , we see that the five solutions are:

$$z = 0, x = 3, y = 0; \quad z = 1, x = -1 = 4, y = -1 = 4; \quad z = 2, x = 0, y = 3; \quad z = 3, x = 1, y = 2; \quad z = 4, x = 2, y = 1.$$

- (5) Compute the real and imaginary parts of $\frac{\pi+i}{5-i} \in \mathbb{C}$.

Solution. Let us compute:

$$\begin{aligned} \frac{\pi+i}{5-i} &= \frac{(\pi+i)(5+i)}{(5-i)(5+i)} \\ &= \frac{(5\pi-1) + (\pi+5)i}{5^2+1^2} \\ &= \frac{5\pi-1}{26} + \frac{\pi+5}{26} \cdot i. \end{aligned}$$

Therefore, the real part is $\frac{5\pi-1}{26}$, while the imaginary part is $\frac{\pi+5}{26}$.

- (6) Fix an integer $n \geq 2$ and set $\zeta = e^{2\pi i/n} \in \mathbb{C}$.

- (a) Show that the product $\zeta \cdot \zeta \cdots \zeta^{n-1}$ is equal to ± 1 . When is it 1 and when is it -1 ?

Solution. Since $\zeta^k = (e^{2\pi i/n})^k = e^{2\pi ki/n}$, this product is simply

$$\prod_{k=1}^{n-1} e^{2\pi ki/n} = e^{\sum_{k=1}^{n-1} 2\pi ki/n},$$

where the sum is

$$\sum_{k=1}^{n-1} 2\pi ki/n = \frac{2\pi i}{n} \sum_{k=1}^{n-1} k = \frac{2\pi i}{n} \cdot \frac{(n-1)n}{2} = \pi i(n-1).$$

Here, in the penultimate equality, we are using the identity

$$\sum_{k=1}^{n-1} k = \frac{n(n-1)}{2}$$

that we showed in Lecture 1.

Therefore, the product we want to compute is equal to

$$e^{(n-1)\pi i}.$$

This is equal to ± 1 depending on the parity of n : If n is *odd*, then $n - 1$ is even, and the angle $(n - 1)\pi$ is the same as 0, so we get 1. If n is even, then $n - 1$ is odd, and the angle is the same as π , which gives us -1 .

- (b) Show that $1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1} = 0$.

Solution. This requires the algebra identity

$$(1 + x + \cdots + x^{n-1})(1 - x) = 1 - x^n.$$

We proved this using induction in Lecture 2. Plugging ζ into this gives us

$$(1 + \zeta + \cdots + \zeta^{n-1})(1 - \zeta) = 1 - \zeta^n = 1 - e^{2\pi i} = 1 - 1 = 0.$$

On the other hand, since $\mathbb{C}$ is a field, we know by 3(c) that one of the factors on the left hand side must be 0. Since $1 \neq \zeta$, this must mean that the first factor is 0, as desired.